

Research Notes on Mixture of experts

Mixture of experts

>> Section 1

Further reading Before deep learning era McLachlan, Geoffrey J.; Peel, David (2000). Finite mixture models. Wiley series in probability and statistics applied probability and statistics section. New York Chichester Weinheim Brisbane Singapore Toronto: John Wiley & Sons, Inc. ISBN 978-0-471-00626-8. Yuksel, S. E.; Wilson, J. N.; Gader, P. D. (August 2012). "Twenty Years of Mixture of Experts". *IEEE Transactions on Neural Networks and Learning Systems*. 23 (8): 1177–1193. Bibcode:2012ITNNL..23.1177Y. doi:10.1109/TNNLS.2012.2200299. ISSN 2162-237X. PMID 24807516. S2CID 9922492. Masoudnia, Saeed; Ebrahimpour, Reza (12 May 2012). "Mixture of experts: a literature survey". *Artificial Intelligence Review*. 42 (2): 275–293. doi:10.1007/s10462-012-9338-y. S2CID 3185688. Nguyen, Hien D.; Chamroukhi, Faicel (July 2018). "Practical and theoretical aspects of mixture-of-experts modeling: An overview". *WIREs Data Mining and Knowledge Discovery*. 8 (4) e1246. doi:10.1002/widm.1246. ISSN 1942-4787. S2CID 49301452. Practical techniques for training MoE Transformer models Zoph, Barret; Bello, Irwan; Kumar, Sameer; Du, Nan; Huang, Yanping; Dean, Jeff; Shazeer, Noam; Fedus, William (2022). "ST-MoE: Designing Stable and Transferable Sparse Expert Models". arXiv:2202.08906 [cs.CL]. Muennighoff, Niklas; Soldaini, Luca; Groeneveld, Dirk; Lo, Kyle; Morrison, Jacob; Min, Sewon; Shi, Weijia; Walsh, Pete; Tafjord, Oyvind; Lambert, Nathan; Gu, Yuling; Arora, Shane; Bhagia, Akshita; Schwenk, Dustin; Wadden, David; Wettig, Alexander; Hui, Binyuan; Dettmers, Tim; Kiela, Douwe; Farhadi, Ali; Smith, Noah A.; Pang Wei Koh; Singh, Amanpreet; Hajishirzi, Hannaneh (2024). "OLMoE: Open Mixture-of-Experts Language Models". arXiv:2409.02060 [cs.CL]., with associated data release at "allenai/OLMoE". Ai2. 2024-10-17. Retrieved 2024-10-18. Rajbhandari, Samyam; Li, Conglong; Yao, Zhewei; Zhang, Minjia; Reza Yazdani Aminabadi; Ammar Ahmad Awan; Rasley, Jeff; He, Yuxiong (2022). "DeepSpeed-MoE: Advancing Mixture-of-Experts Inference and Training to Power Next-Generation AI Scale". arXiv:2201.05596 [cs.LG]. DeepSeek-AI; et al. (2024). "DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model". arXiv:2405.04434 [cs.CL]. DeepSeek-AI; et al. (2024). "DeepSeek-V3 Technical Report". arXiv:2412.19437 [cs.CL]. Jin, Chao; Jiang, Ziheng; Bai, Zhihao; Zhong, Zheng; Liu, Juncai; Li, Xiang; Zheng, Ningxin; Wang, Xi; Xie, Cong; Huang, Qi; Heng, Wen; Ma, Yiyuan; Bao, Wenlei; Zheng, Size; Peng, Yanghua; Lin, Haibin; Liu, Xuanzhe; Jin, Xin; Liu, Xin (2025). "MegaScale-MoE: Large-Scale Communication-Efficient

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>> Section 2

Mixture of experts (MoE) is a machine learning technique where multiple expert networks (learners) are used to divide a problem space into homogeneous regions. MoE represents a form of ensemble learning. They were also called committee machines.

>> Section 3

Capacity factor Suppose there are n experts in a layer. For a given batch of queries $\{x_1, x_2, \dots, x_T\}$, each query is routed to one or more experts. For example, if each query is routed to one expert as in Switch Transformers, and if the experts are load-balanced, then each expert should expect on average T/n queries in a batch. In practice, the experts cannot expect perfect load balancing: in some batches, one expert might be underworked, while in other batches, it would be overworked. Since the inputs cannot move through the layer until every expert in the layer has finished the queries it is assigned, load balancing is important. The capacity factor is sometimes used to enforce a hard constraint on load balancing. Each expert is only allowed to process up to $c \cdot T/n$ queries in a batch. The ST-MoE report found $c \in [1.25, 2]$ to work well in practice.

>> Section 4

Routing In the original sparsely-gated MoE, only the top-k experts are queried, and their outputs are weighted-summed. There are other methods. Generally speaking, routing is an assignment problem: How to assign tokens to experts, such that a variety of constraints are followed (such as throughput, load balancing, etc.)? There are typically three classes of routing algorithm: the experts choose the tokens ("expert choice"), the tokens choose the experts (the original sparsely-gated MoE), and a global

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assigner matching experts and tokens. During inference, the MoE works over a large batch of tokens at any time. If the tokens were to choose the experts, then some experts might get few tokens, while a few experts get so many tokens that it exceeds their maximum batch size, so they would have to ignore some of the tokens. Similarly, if the experts were to choose the tokens, then some tokens might not be picked by any expert. This is the "token drop" problem. Dropping a token is not necessarily a serious problem, since in Transformers, due to residual connections, if a token is "dropped", it does not disappear. Instead, its vector representation simply passes through the feedforward layer without change. Other approaches include solving it as a constrained linear programming problem, using reinforcement learning to train the routing algorithm (since picking an expert is a discrete action, like in RL). The token-expert match may involve no learning ("static routing"): It can be done by a deterministic hash function or a random number generator.

>> Section 5

Adaptive mixtures of local experts The adaptive mixtures of local experts uses a Gaussian mixture model. Each expert simply predicts a Gaussian distribution, and totally ignores the input. Specifically, the i -th expert predicts that the output is $y \sim N(\mu_i, I)$, where μ_i is a learnable parameter. The weighting function is a linear-softmax function: $w_i(x) = \frac{e^{k_i^T x + b_i}}{\sum_j e^{k_j^T x + b_j}}$. The mixture of experts predict that the output is distributed according to the log-probability density function: $\ln f_\theta(y|x) = \ln \left[\sum_i e^{k_i^T x + b_i} \sum_j e^{k_j^T x + b_j} N(y|\mu_i, I) \right] = \ln \left[(2\pi)^{-d/2} \sum_i e^{k_i^T x + b_i} \sum_j e^{k_j^T x + b_j} e^{-\frac{1}{2} \|y - \mu_i\|^2} \right]$. It is trained by maximal likelihood estimation, that is, gradient ascent on $f(y|x)$. The gradient for the i -th expert is